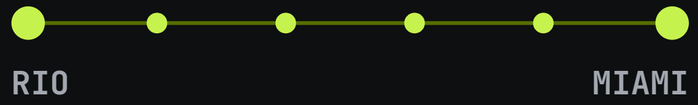
TESTNET · TEST KEYS · NOT REAL CUSTODIANS

# Radox Custody Rail

**One record for the goods, from origin to delivery, that every company along the way could share.**

Scan: the live shipment
rio-miami-rail.vercel.app

Toward a SWIFT-like standard on Stellar · Pilot corridor: Rio–Miami




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PMF · THE PROBLEM

# Who holds the whole chain?

Warehouse — Port — Carrier — Customs — Recipient

**Our hypothesis:** each company records only its own leg; no shared record follows the goods. Not yet tested with a customer.

## 45M

**ocean bills of lading issued a year**

DCSA, Feb 2023 (2021 data)

## ~11%

**of bills of lading issued electronically, 2025**

DCSA, as of 2025 (Jan 2025 guide: 5.7%)

## 22%

**of recorded cargo thefts in 2025 involved insiders**

TT Club & BSI Cargo Theft Report, Apr 2026

Context: our explainer video about a Rio–Miami trade-corridor initiative poses one possibility: a unique digital record for every shipment that both sides can check. This project is a custody layer for that record.

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PMF · WHAT IT COSTS

# Cargo goes missing. A bank got fooled.

**Our hypothesis:** a financier cannot see who holds the goods before paying. No customer has confirmed it yet.

## R$900M

**est. direct loss, 8,570 cargo robberies in Brazil, 2025**

NTC&Logística survey, 28 Mar 2026

## US$85M

**HSBC's loss on forged documents for oil trades that never happened**

Hin Leong case, Global Trade Review, 18 Nov 2024

## US$2.5T

**global trade finance gap (market size only)**

ADB survey of banks, Dec 2025

These numbers size the market problem. They are not results of ours, and we do not claim our build prevents any of it.

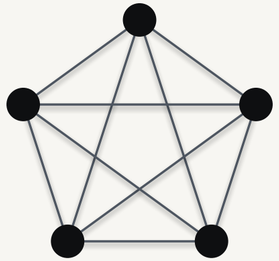
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THE PROPOSAL · A PUBLIC STANDARD

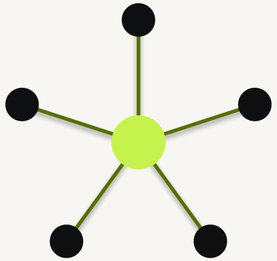
# A SWIFT-like standard: integrate once

**The goal is** a shared, SWIFT-like custody standard that companies along a supply chain, from origin to destination, could use. **Neither the standard nor the rail exists yet.**

PAIR BY PAIR · 10 LINKS



ONE STANDARD · 5 LINKS



Example, 5 companies: 10 custom links, or 5 links to one standard.

OPEN · ANYONE

— Record format, roles and events: the custody_registry contract (Apache-2.0)

— Verify any shipment, no login (slide 6)

RADOX OPERATES · CLIENT PAYS

— Onboarding, issuance controls, operation of the rail

— The API and backend: closed, not self-hostable

— Subscription per connected institution + fee per shipment; no prices yet (slide 9)

Each company would integrate once instead of pair by pair, but would still need a key and one connection. No port, airport or chamber link: none confirmed. Radox is the rail, never the issuer. Regulation and licences stay with the client and its local counsel.

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THE RAIL · DESIGN, LIFECYCLE BUILT

# One rail, origin to delivery

One record would follow the goods across company boundaries; today it runs across five test keys.




— **Rule of every link:** both sides sign one Stellar transaction (delivery: recipient and holder), or the call fails.
— Each step extends the on-chain `head`: anyone can replay the history from Horizon and check it against the head.
— The token stays with its holder in transit; only custody moves.

**Built: this lifecycle on testnet, test keys. Design, not built: real companies on the rail, the standard, integrations.**

Keyed by sha256(container | origin); would sit beside the carrier's bill of lading, not integrated. [1] Radox policy, not enforced on-chain; redeem needs the holder's signature.

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THE PROOF · LIVE ON TESTNET

# The lifecycle ran on testnet

| | |
|---|---|
| Warehouse → Port, both signed | ledger 5381142 · 635c99…f0bd38 |
| Port → Carrier, both signed | ledger 5381143 · 56a0f0…7b7cc2 |
| Event attested by Source | ledger 5381144 · 333792…068905 |
| Carrier → Customs, both signed | ledger 5381145 · ac268a…c3c4b2 |
| Customs → Recipient, both signed | ledger 5381146 · 5199d4…400b7a |
| Delivered by Recipient to Holder, both signed | ledger 5381147 · 919def…144e37 |
| Redeemed: token burned | 864fd1…15644d |

Verify head

`2b3d51f0` open · operator
`635c99ac` handover 1 · Warehouse → Port
`56a0f006` handover 2 · Port → Carrier
`ac268a49` handover 3 · Carrier → Customs
`5199d4fb` handover 4 · Customs → Recipient
`919def8c` deliver · Recipient → Holder
`864fd164` burn · existing redeem

**7** successful

**9** refused, e.g. wrong signer (simulated)

**24** unit tests

**MATCH** head check, 25 Oct

`custody_registry`
`CCBJVXMJYWPHTDJHDFPRGKPGXGEP24JOPJAMUS5C3BMZJTRTWR2H4LII`

Second shipment (proofs/three-machine.json): two team members each signed their handovers from their own computers and keys; still test keys.

**All test keys: this proves the mechanism, not independent companies.**

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ECOSYSTEM · WHY STELLAR

# Why Stellar: all signers, one call

| | | |
|---|---|---|
| 01 | **Multi-party auth in one call** | Soroban authorises both companies of a link inside one contract invocation: the handover is atomic. |
| 02 | **History from Horizon, checked by an on-chain head** | RPC keeps contract events about 7 days, so we replay each shipment's transactions from Horizon and check them against the head stored in the contract. |
| 03 | **The token stays put, then burns** | The shipment token never moves in transit; the existing Radox redeem burns it after delivery. |

**Nearest neighbours we found on Stellar** (LumenLoop, SCF, Scout, 24–25 Oct): Kutana and Trustless Work release escrow on trade milestones; Pipeline finances commodity trade. We found none that records a custody handover signed by two custodians, and no SEP.

**Open source, Apache-2.0:** a custody primitive other Stellar real-world-asset issuers can reuse.

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HACKATHON PROGRESS

# Built this weekend, on disclosed prior work

## PRIOR WORK · BEFORE SUN 25 OCT 09:00

— Radox platform (closed backend); its 3 Soroban contracts are on Stellar mainnet, public since 2026-08-19

— transfer_agent and its redeem (new testnet instance from the existing wasm)

— Architecture notes (20 Oct), planning files (24–25 Oct), the explainer video, our agent tooling (from 22 Oct)

— No contract code before then: first repo commit 25 Oct 09:00 UTC

## BUILT AT HACKMERIDIAN

— custody_registry contract, 24 unit tests

— Lifecycle harness: 7 testnet transactions, 9 refused calls

— verify-head script, the public shipment page, this deck

— Radox API reads the registry: same head (closed backend, testnet)

**AI attribution** Code, tests, scripts, the shipment page, docs and this deck were made with AI coding agents (Claude Code); the team decided scope and what ships. Per-file record in AI_USAGE.md. The video narration is an AI voice generated from Pedro's voice.

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MARKET · MODEL · THE ASK

# One rail, every corridor

## MARKET · FIGURES, NOT OURS

**US$94.2B**

**US–Brazil goods trade, 2025[1]**

**US$6.1B**

**Florida's goods exports to Brazil, its #1 market, 2025[2]**

**Global:** US$26.26T world merchandise exports, 2025[3]; ~80% of goods trade by volume goes by sea[4]; ~45M ocean bills of lading a year[5].

## MODEL · NO PRICES YET

**Subscription per connected institution + a fee per shipment.**

Borrows the shape of Swift's billing: annual charges, plus traffic fees for messages sent[6]. Not affiliated with Swift.

**Now** testnet mechanism; no standard, users or revenue. **Next** one Rio–Miami pilot shipment (none agreed); SCF Build, audit, mainnet; more corridors.

[1] US Census Bureau, Trade in Goods with Brazil, 2025 (exports + imports, sum ours) [2] USTR, Florida trade data [3] WTO, Mar 2026 [4] UNCTAD, Review of Maritime Transport [5] DCSA, Feb 2023 [6] swift.com billing rules. Market figures, not Radox volumes; no traction implied.

**Our ask** **Introductions to one financial institution and one port, airport or logistics operator for a first pilot shipment, and a Stellar engineer's review of the design.**

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THE TEAM

# Why us: a founder already on Stellar mainnet

## Pedro Wakigawa Saragossy

Founder, captain. Built Radox: 3 Soroban contracts on Stellar mainnet.

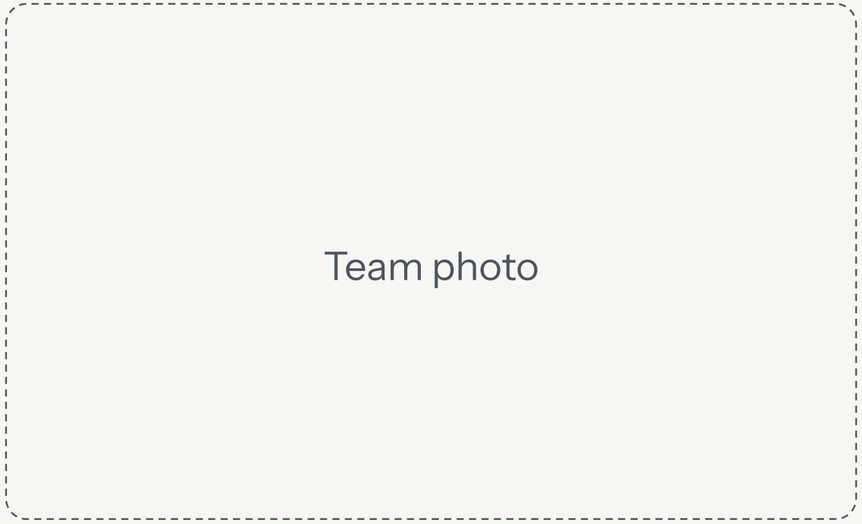
## Leo Schlanger

Developer: custody_registry, the testnet harness and the proofs.

## Vhendala

QA and product review: reads every deliverable as a judge would.

Leo and Vhendala joined for the hackathon.




**Rio–Miami first. Our goal: one rail for goods, every corridor.**

**Live page** rio-miami-rail.vercel.app **Code** github.com/pwsaragossy/rio-miami-rail (Apache-2.0)

**AI** Code, page, docs and deck made with AI coding agents (Claude Code). Narration: an AI voice from Pedro's voice. No AI-generated images in this deck.

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